

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey**

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 1 – Research Paper Review / Literature Survey
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	RASHMI NAURENE	Reg. No.:	192521357
Date:	22.08.2026	Duration:	60 minutes

Code of Conduct

I RASHMI NAURENE (192521357) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Rashmi

Annexure A – Research Paper Review / Literature Survey**Instructions to Students:**

Each student is assigned an individual problem statement. Based on the assigned problem statement, **conduct a literature survey and review relevant research papers** to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select **relevant research papers** related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the **objectives, methodologies, technologies, datasets, and major findings** of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as **accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics**.
4. Identify the **limitations and challenges** of the existing research approaches.
5. Analyze the literature to identify the **research gap** or unresolved issues related to the assigned problem.
6. Discuss the **possible improvements or future research directions** based on the identified research gap.
7. Prepare a **comparative literature survey table** containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper **citations and references** using a consistent referencing style.

Deliverable: Submit a **Research Paper Review / Literature Survey Report** containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

1 · Introduction & Objective of the Survey

The assigned problem, the Employee Skill-Gap Analysis and Training Recommendation System, sits at the intersection of three active research areas: automated skill-gap detection, personalized training/course recommendation, and AI-driven competency management. This survey reviews seven papers published between 2023 and 2026 that address one or more of these areas, in order to understand existing approaches, technologies, and datasets; compare their methodologies and reported performance; identify their limitations; and locate the specific research gap the assigned system is positioned to address.

1.1 Survey Scope

- Automated skill-gap identification from resumes, performance data, or employee records (Papers 2, 4, 5).
- AI/ML-driven competency mapping and workforce development strategy (Papers 3, 6).
- Personalized training/course/content recommendation systems, including cold-start handling (Papers 1, 7).

2 · Selected Research Papers

Seven papers were selected based on direct relevance to skill-gap analysis, competency mapping, or training/course recommendation, all published within the last three years (2023–2026).

Paper 1 — Personalized E-Learning Recommender System Based on Autoencoders

El Youbi El Idrissi, L., Akharraz, I., & Ahaitouf, A. (2023). Applied System Innovation, 6(6), 102.

Objective: Personalize e-learning content recommendations for individual learners, addressing the high dimensionality and sparsity typical of learner-course interaction data. **Methodology:** An autoencoder (deep learning) is used for dimensionality reduction and feature extraction on top of a collaborative-filtering recommendation approach. **Dataset/Tools:** The e-learning interaction dataset originally created by Kulkarni et al.; implemented using TensorFlow and Keras. **Key Findings:** The autoencoder-based collaborative filtering model outperformed KNN, SVD, SVD++, and NMF baselines on RMSE and MAE. **Limitations:** Built for general e-learning platforms rather than a workplace competency context; does not incorporate role-based skill requirements or manager oversight.

Paper 2 — Placement Prediction and Skill Gap Analysis Using Machine Learning Model

Vijaykumar, M., Vinutha, M. R., et al. (2024). Proc. International Conference on Green Computing for Communication Technologies (ICGCCT 2024).

Objective: Predict candidate placement outcomes and analyze skill gaps relative to industry/placement requirements. **Methodology:** Machine learning-based predictive modelling applied to academic and placement records. **Dataset/Tools:** Institutional academic and placement history records. **Key Findings:** ML-based models can flag skill deficiencies and estimate placement readiness ahead of recruitment cycles. **Limitations:** Oriented toward one-time academic placement prediction rather than continuous, on-the-job competency tracking or ongoing training recommendation.

Paper 3 — Machine Learning and AI Technology-Induced Skill Gaps and Opportunities for Continuous Development of Middle-Skilled Employees

Chuang, S., Shahhosseini, M., Javaid, M., & Wang, G. G. (2024). *Journal of Work-Applied Management (Emerald)*. DOI: 10.1108/JWAM-08-2024-0111

Objective: Examine how the adoption of ML/AI in workplaces itself creates new skill gaps for middle-skilled employees, and propose strategic upskilling/reskilling interventions. **Methodology:** Conceptual and empirical HRD research synthesis with scenario analysis, informed by a PRISMA-style review approach. **Dataset/Tools:** Synthesis of existing HRD/workforce-development literature; no single proprietary dataset. **Key Findings:** Proposes proactive, AI-integrated career-development interventions; flags the risk that AI-driven workforce tools can unintentionally reinforce bias. **Limitations:** Primarily conceptual and policy-oriented; does not present a deployable skill-gap detection or recommendation system.

Paper 4 — Optimizing Workforce Development through Machine Learning: A Comprehensive Analysis of Skill Gaps and Predictive Insights

Gayathri, K. C., et al. (2025). *IEEE Conference Publication*.

Objective: Use machine learning to identify and address employee skill gaps and optimize training/talent-management decisions. **Methodology:** Comparative application of decision trees, random forest, and neural networks to employee performance metrics and training records, validated with cross-validation. **Dataset/Tools:** Employee performance metrics and training records (organizational dataset). **Key Findings:** Random forest and neural networks achieved strong performance in forecasting skill demand at scale; decision trees offered more interpretable results. **Limitations:** Focused on model comparison and prediction accuracy rather than end-to-end delivery of personalized training recommendations to employees.

Paper 5 — Skill Gap Analysis Using Machine Learning

Rameshbabu, V., et al. (2025). *International Journal of Engineering Research and Sustainable Technologies (IJERST)*.

Objective: Bridge the gap between a candidate's existing skillset and industry requirements using resume-based analysis. **Methodology:** Natural Language Processing (NLP) and machine learning applied to resume text to detect skill gaps and generate a personalized improvement roadmap. **Dataset/Tools:** User resumes; NLP pipelines for text/skill extraction. **Key Findings:** The system produces a personalized development roadmap and recommends interview-preparation resources aligned to the user's chosen professional domain. **Limitations:** Relies solely on resume text as input; does not incorporate ongoing performance reviews, assessments, or project-experience data as the assigned problem statement requires.

Paper 6 — Utilizing Artificial Intelligence for Competency Mapping and Personalised Skill Development in IT Organizations

Talodhikar, S., & Farooqui, S. (2025). *Journal of Information Systems Engineering and Management (JISEM)*, 10(15s).

Objective: Apply AI to map employee competencies against role requirements and personalize skill-development plans within IT organizations. **Methodology:** Framework-based approach: determining employee competencies and proficiency levels, selecting training courses to close identified gaps, and evaluating training-program effectiveness. **Dataset/Tools:** IT-organization competency and training-program data (case/framework discussion). **Key Findings:** AI-driven competency mapping is argued to support a more agile, skilled, and competitive workforce when applied responsibly. **Limitations:** Presented largely at the framework/discussion level, with limited quantitative benchmarking against alternative competency-mapping methods; scoped to IT organizations specifically.

Paper 7 — Robust Recommendation for Personalized Training with Deepfake-Assisted Generation Models

Zhu, X., Xie, Y., & Fakhrabadi, A. K. (2026). *Complex & Intelligent Systems*, 12, Article 157. DOI: 10.1007/s40747-026-02300-9

Objective: Improve the robustness of personalized training recommendations under data sparsity and cold-start conditions. **Methodology:** Similarity-based synthetic (deepfake-inspired) data generation combined with hybrid collaborative and content-based filtering; evaluated with HR@K, NDCG@K, Diversity, and Coverage metrics. **Dataset/Tools:** MovieLens-1M benchmark dataset, adapted to a training-recommendation evaluation setting. **Key Findings:** The proposed hybrid method consistently outperformed CF, CBF, a standard hybrid recommender, and a deep-learning recommender, with the largest gains in cold-start scenarios. **Limitations:** Evaluated on a general-purpose benchmark dataset rather than an HR/competency-specific one; does not model organizational role-competency frameworks directly.

3 · Comparative Literature Survey Table

Author/Y ear	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
El Youbi El Idrissi et al., 2023	E-learning content personalization	Autoencoder + collaborative filtering	Kulkarni et al. dataset; TensorFlow/ Keras	Outperforms KNN/SVD/SVD++/NMF on RMSE, MAE	No role-based competency context	No link to job-role skill requirements
Vijaykumar et al., 2024	Placement & skill-gap prediction	ML-based predictive modelling	Institutional academic/place ment records	Identifies placement readiness and gaps	One-time academic scope only	No continuous, on-the-job tracking
Chuang et al., 2024	AI-induced skill gaps in middle-skilled workers	PRISMA-informed HRD synthesis + scenario analysis	Literature synthesis	Proposes AI-integrated upskilling interventions	Conceptual, not a deployable system	No implemented gap-detection engine
Gayathri et al., 2025	Workforce skill-gap prediction	Decision trees, random forest, neural networks	Employee performance & training records	RF/NN strong on large-scale forecasting	No recommendation delivery layer	Prediction not connected to training recs
Rameshbabu et al., 2025	Resume-based skill-gap detection	NLP + ML on resume text	User resumes	Personalized roadmap + interview resources	Resume-only input	Ignores performance/assessment/project data
Talodhikar &	AI competenc	Competency-mapping framework	IT-org competency/training data	Supports agile, skilled workforce	Framework-level, limited	Sector-specific, not

Farooqui, 2025	y mapping (IT sector)				benchmarking	generalized/quantified
Zhu, Xie & Fakhrabadi, 2026	Cold-start robustness in training recommendation	Synthetic-data augmentation + hybrid filtering	MovieLens-1M (adapted)	Outperforms CF/CBF/Hybrid/DLR baselines	Generic benchmark, not HR-specific	No organizational competency-framework model

4 · Comparative Analysis

4.1 Accuracy and Performance

Among the recommendation-focused papers, deep-learning and hybrid approaches consistently outperform simpler collaborative- or content-based baselines. El Youbi El Idrissi et al. (2023) report autoencoder-based filtering beating KNN/SVD-family baselines on RMSE and MAE, and Zhu et al. (2026) report their synthetic-data-augmented hybrid model beating CF, CBF, a standard hybrid recommender, and a deep-learning recommender across HR@K, NDCG@K, diversity, and coverage. On the prediction side, Gayathri et al. (2025) found ensemble methods (random forest, neural networks) more effective than single decision trees for large-scale skill-demand forecasting, at the cost of interpretability and training time.

4.2 Scalability

Zhu et al. (2026) explicitly report training-time trade-offs: their hybrid method scales better than a pure deep-learning recommender while still benefiting from synthetic-data augmentation, whereas Gayathri et al. (2025) note that neural-network-based forecasting becomes markedly more computationally expensive as dataset size and augmentation levels grow. None of the papers reviewed test at an enterprise-wide HR scale (tens of thousands of employees), which is a practical scalability question the assigned system will need to answer directly.

4.3 Methodology

Methodologies cluster into three families: (i) collaborative/content-based/hybrid filtering with deep-learning enhancements for recommendation quality (Papers 1, 7); (ii) supervised ML classification/regression for skill-gap or placement prediction (Papers 2, 4); and (iii) framework-level or conceptual HRD/competency-mapping approaches without a fully implemented pipeline (Papers 3, 6). Paper 5 sits between families (i) and (ii), applying NLP feature extraction ahead of a recommendation-style output.

4.4 Efficiency & Practical Deployment

Papers reporting concrete systems (1, 5, 7) all note practical constraints — dimensionality/sparsity (Paper 1), reliance on a single input source such as resumes (Paper 5), and the need for synthetic data specifically to remain efficient under sparse real-world interaction volumes (Paper 7) — indicating that data availability, not algorithmic sophistication alone, is often the binding constraint on deployability.

5 · Limitations and Challenges of Existing Approaches

- Single-source input: most systems reviewed rely on one data source — resumes (Paper 5), generic interaction logs (Papers 1, 7), or performance records alone (Paper 4) — rather than combining performance reviews, certifications, assessments, and project experience as the assigned problem statement requires.
- Disconnection between prediction and recommendation: skill-gap prediction (Papers 2, 4) and training recommendation (Papers 1, 7) are treated as separate research problems in the papers reviewed, with no single pipeline connecting a detected gap directly to a ranked, actionable training suggestion.
- Lack of organizational competency-framework modelling: none of the seven papers explicitly models department-specific or role-specific competency frameworks against which an individual's skills are scored, despite this being central to the assigned problem statement.
- Conceptual-only contributions: Papers 3 and 6 propose strategic or framework-level approaches without a benchmarked, implemented system, leaving open how such frameworks would perform quantitatively.
- Cold-start and data sparsity: even the most technically advanced recommendation paper reviewed (Paper 7) treats cold-start as an open challenge requiring synthetic data augmentation, which introduces its own risk of noise at high augmentation levels.
- Evaluation on generic, non-HR datasets: Paper 7's strongest results are demonstrated on MovieLens-1M, a general recommendation benchmark, rather than an HR/competency dataset, leaving a gap in domain-specific empirical validation.

6 · Identified Research Gap

Across the seven papers reviewed, no single system integrates all of the following, which together define the assigned problem statement: (1) multi-source competency assessment combining performance reviews, certifications, assessments, and project experience; (2) automated skill-gap scoring against explicit, role-specific and department-specific competency frameworks; (3) personalized, ranked training recommendations generated directly from the detected gap; (4) manager- and HR-facing dashboards and analytical reporting; and (5) automated notifications and learning-path tracking to drive completion. Existing work addresses at most two or three of these components in isolation — for example, resume-based gap detection without ongoing performance data (Paper 5), or robust recommendation without organizational competency modelling (Paper 7). The assigned Employee Skill-Gap Analysis and Training Recommendation System is positioned precisely to close this integration gap rather than to outperform any single component in isolation.

7 · Possible Improvements & Future Research Directions

- Combine the multi-source input handling implied by Papers 2, 4, and 5 (academic/performance/resume data) into a single unified employee competency profile, rather than treating each source as a separate research problem.
- Adopt the synthetic-data augmentation strategy from Paper 7 specifically to address cold-start for newly onboarded employees or newly defined roles, where historical training-interaction data does not yet exist.
- Operationalize the conceptual competency-mapping frameworks proposed in Papers 3 and 6 into an explicit, queryable role-competency data model that the skill-gap engine can compare an employee profile against directly.
- Extend evaluation practice beyond generic benchmarks (as in Paper 7) toward HR-specific datasets or organizationally realistic simulated data, so that recommendation-quality metrics (HR@K, NDCG@K) are validated in the actual target domain.

- Investigate treeification-style worst-case guarantees (as used in production hash-table implementations) as an analogy for bounding worst-case latency in the gap-analysis engine as organizational headcount scales.
- Incorporate the bias-risk caution raised by Chuang et al. (2024) into the system's design from the outset — auditing recommendation and grading logic for unintended bias across demographic groups rather than treating fairness as an afterthought.

8 · References

References are listed in IEEE numbered style, matching their first citation order in this survey.

- [1] L. El Youbi El Idrissi, I. Akharraz, and A. Ahaitouf, “Personalized E-Learning Recommender System Based on Autoencoders,” *Applied System Innovation*, vol. 6, no. 6, p. 102, 2023, doi: 10.3390/asi6060102.
- [2] M. Vijaykumar, M. R. Vinutha, et al., “Placement Prediction and Skill Gap Analysis Using Machine Learning Model,” in *Proc. Int. Conf. Green Computing for Communication Technologies (ICGCCT)*, 2024.
- [3] S. Chuang, M. Shahhosseini, M. Javaid, and G. G. Wang, “Machine Learning and AI Technology-Induced Skill Gaps and Opportunities for Continuous Development of Middle-Skilled Employees,” *Journal of Work-Applied Management*, 2024, doi: 10.1108/JWAM-08-2024-0111.
- [4] K. C. Gayathri et al., “Optimizing Workforce Development through Machine Learning: A Comprehensive Analysis of Skill Gaps and Predictive Insights,” *IEEE Conference Publication*, 2025.
- [5] V. Rameshbabu et al., “Skill Gap Analysis Using Machine Learning,” *International Journal of Engineering Research and Sustainable Technologies (IJERST)*, 2025.
- [6] S. Talodhikar and S. Farooqui, “Utilizing Artificial Intelligence for Competency Mapping and Personalised Skill Development in IT Organizations,” *Journal of Information Systems Engineering and Management (JISEM)*, vol. 10, no. 15s, 2025.
- [7] X. Zhu, Y. Xie, and A. K. Fakhrabadi, “Robust Recommendation for Personalized Training with Deepfake-Assisted Generation Models,” *Complex & Intelligent Systems*, vol. 12, art. 157, 2026, doi: 10.1007/s40747-026-02300-9.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies, tools, datasets, results, and performance using a clear comparison table	Provides a good comparison with relevant parameters	Provides limited comparison with few parameters	Little or no meaningful comparison
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	Mark
Selection & Review of Research Papers(25%)	/5
Comparative Analysis(20%)	/5
Critical Analysis & Research Gap(25%)	/5
Future Directions & Relevance(15%)	/5
Documentation, Citations & References(15%)	/5
TOTAL	/25